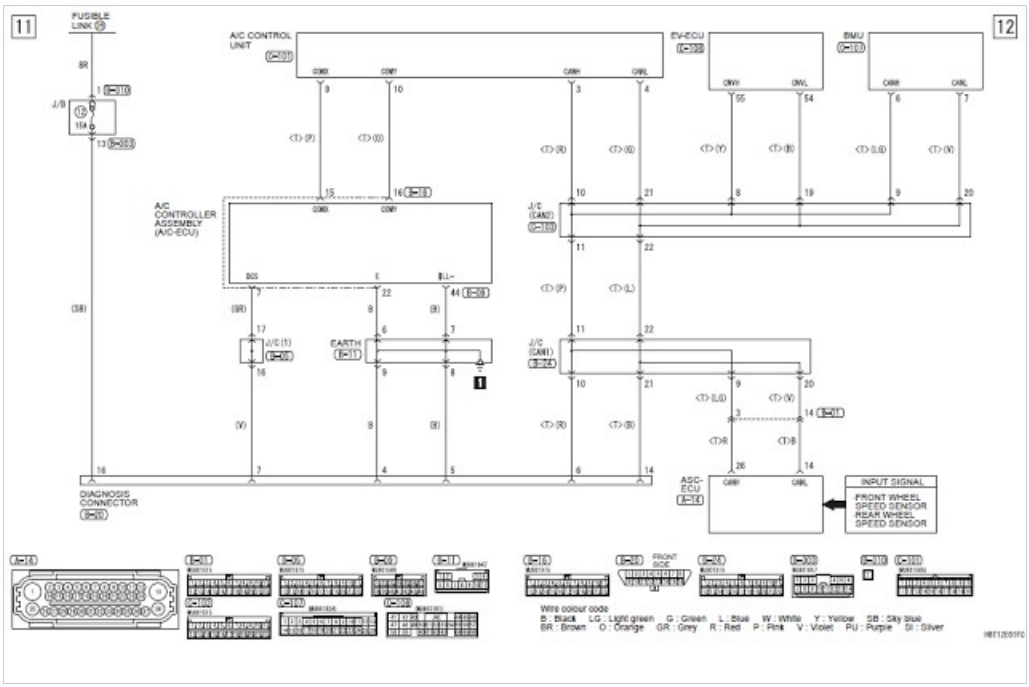
I have been able to capture the status messages being sent from the compressor. using an AVR-CAN spying on the network with only the compressor connected. The message sent is from node ID 0x439.

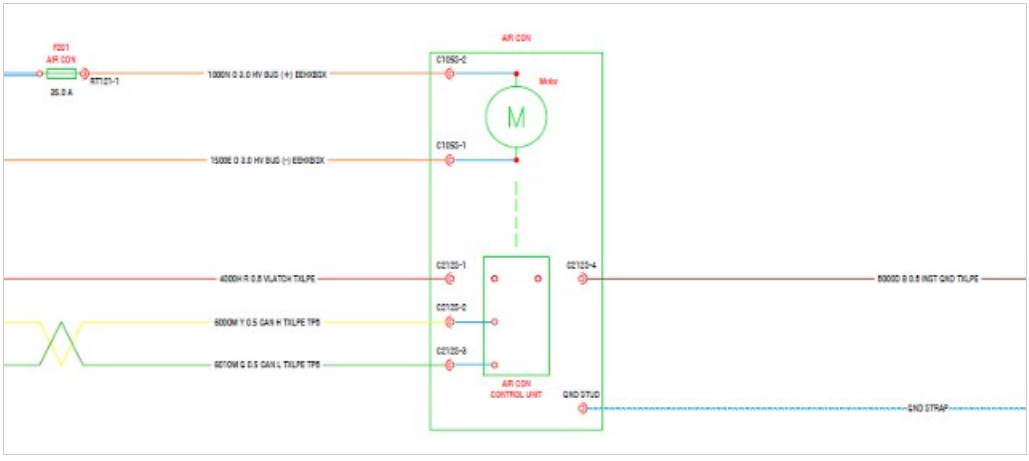
The AC compressor send the following messages:
439 0300000000000000 when off and
439 0000000000000000 when on.

Below are the schematics and a few photos. The PCBs are all immersed in a gelatinous goo. Messy stuff. This is probably to increase the insulation between high voltage points, and protect the HV circuits from humidity and moisture.

From the photos it can be seen the HV +ve is Orange with a white stripe. -ve is Orange with a Black stripe.



i-MiEV



AZD Schematic

(1) Electric driven compressor

Table 2.1 Electric driven compressor spec.

L * W * H	291 * 162 * 157 (mm)
Mass	10.2kg (with Bracket)
Compressor type	Scroll with rare earth metal motor
Displacement	30 cc/rev.
Inverter	Integrated, suction ref. cooling
Max. rev.	6000 rpm
High Voltage range	DC 220 ~ 400 V
Low Voltage range	DC 8 ~ 16 V
Max. power	4.5 kW
Max. input current	20.5 A (@DC 220V)
Refrigerant	HFC-134a
Ref. Lubricant	POE oil

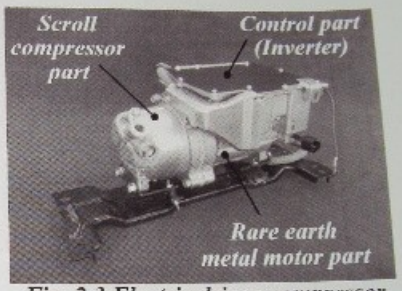


Fig. 2.3 Electric driven compressor

Oil separator is integrated due to improvement of both ref cycle capacity and efficiency.

SAE International